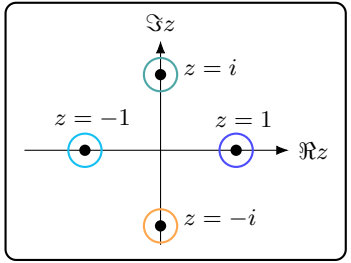
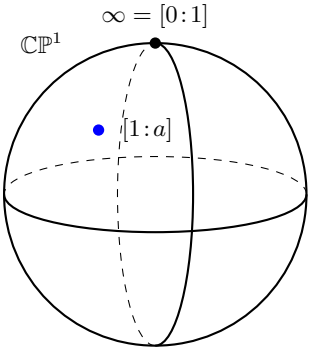
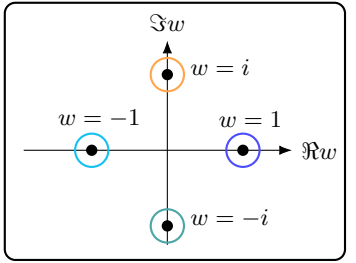
$z - a$ has a simple zero at $[1:a]$
$\operatorname{ord}_{[1:a]}(z-a) = +1$



$U_0 = \{[Z_0 : Z_1] \mid Z_0 \neq 0\}$ with coordinate $z = \dfrac{Z_1}{Z_0}$



$\frac{1-aw}{w}$ has a simple pole at $[w:1]$
$\operatorname{ord}_{[w:1]}(\frac{1-aw}{w}) = 1$



$U_\infty = \{[Z_0 : Z_1] Z_1 \neq 0\}$ with coordinate $w = \dfrac{1}{z} = \dfrac{Z_0}{Z_1}$